

Yudong Hu

+86-13332836617 |  eudon.hu@gmail.com |  abdn-hyd.github.io
 Yudong (Shawn) Hu |  abdn-hyd

EDUCATION

- **University of Aberdeen** Sep. 2021 - Jun. 2025
 - Bachelor of Science in Artificial Intelligence (First-Class Honours)
 - GPA: 20.06/22.00
- **South China Normal University** Sep. 2021 - Jun. 2025
 - Bachelor of Engineering in Artificial Intelligence
 - GPA: 88.0/100.0

RESEARCH INTERESTS

Interpretable Multimodal & Vision Backbones | Efficient LLMs | Trustworthy AI

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [S.1] Suzhong Fu, Jingqi Dong, **Yudong Hu**, Jing Wang, Xuan Ding, Lunhong Bao, Rui Sun, Yizhou Yang, Yiming Yang, Qi Dou, Shuguang Cui, and Zhen Li (2026). ERA-CA: Evidence-to-Reasoning Architecture for Coronary Angiography. Submitted to *Transactions on Medical Imaging*.
- [C.1] **Yudong Hu**, Yueju Han, Rui Sun, and Jinke Ren. (2026). *MSPCaps: A Multi-Scale Patchify Capsule Network with Cross-Agreement Routing for Visual Recognition*. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 40(6), 4968-4975. DOI: [10.1609/aaai.v40i6.42501](https://doi.org/10.1609/aaai.v40i6.42501).
- [J.1] Huijun Xing, Rui Sun, Jinke Ren, Jun Wei, Chun-Mei Feng, Xuan Ding, Zilu Guo, Yu Wang, **Yudong Hu**, Wei Wei, Xiaohua Ban, Chuanlong Xie, Yu Tan, Xian Liu, Shuguang Cui, Xiaohui Duan, and Zhen Li. (2025). *FlexFair: Achieving Flexible Fairness Metrics in Federated Learning for Medical Image Analysis*. *Nature Communications*, 16, 3342. DOI: [10.1038/s41467-025-58549-0](https://doi.org/10.1038/s41467-025-58549-0).

RESEARCH EXPERIENCES

- **AngioCoT: Interpretable VLM Reasoning for Clinical Diagnostics** Dec. 2025 - Feb. 2026
Research Assistant; The Chinese University of Hong Kong, Shenzhen
Supervisor: Prof. Zhen Li and Dr. Rui Sun
 - Formulated a structured, 10-step data training format and introduced a dynamic step-wise training methodology to enhance the clinical reasoning capabilities of VLMs.
 - Applied DPO to strictly align the model's diagnostic preferences with expert guidelines, significantly improving the interpretability and achieving a 42.3% relative improvement in SYNTAX classification F1 score.
 - Spearheaded the experimental design and evaluation pipeline across critical diagnostic tasks, including SYNTAX classification, thrombus detection, and lesion quantification.
 - Validated the framework on state-of-the-art VLMs (e.g., Qwen-VL, InternVL), demonstrating a 16.78% average increase in F1 score and an 11.82% relative reduction in MAE across all tasks in low-resource medical scenarios.
- **Hessian-Aware Sparse & Low-Rank Quantization for LLMs** Sep. 2025 - Dec. 2025
Research Assistant; The Chinese University of Hong Kong, Shenzhen
Supervisor: Prof. Jinke Ren
 - Developed a hybrid framework to factorize LLM weights into sparse and low-rank components, which isolates outliers from structural redundancies.
 - Constructed a Hessian-weighted objective function that incorporates activation statistics to estimate and minimize layer-wise quantization error.
 - Designed an iterative alternating-minimization algorithm to find the optimal parameter allocation between the k -rank and s -sparse components for partial quantization.
- **Multi-Scale Based Patchify Capsule Network for Visual Recognition** May 2025 - Nov. 2025
Research Assistant; The Chinese University of Hong Kong, Shenzhen 
Supervisor: Prof. Jinke Ren and Dr. Rui Sun
 - Developed a scalable MSPCaps framework (344.3K to 10.9M parameters), generating primary capsules via uniform patchify operations on a ResNet backbone.
 - Proposed an interpretable Cross-Agreement Routing to adaptively match optimal part-to-whole pairs for voting.
 - Achieved state-of-the-art performance across 4 benchmarks (92.88% accuracy on CIFAR-10) and maintained ~80% accuracy against severe FGSM & BIM attacks ($\epsilon = 0.3$).
- **Multimodal Medical Image Prediction & Fairness Analysis** Dec. 2023 - Apr. 2025
Research Assistant; The Chinese University of Hong Kong, Shenzhen 
Supervisor: Prof. Zhen Li, Prof. Xiaohui Duan and Dr. Rui Sun
 - Standardized irregular 3D MRI data by applying multiple resampling techniques (Lanczos, Bilinear, etc.); processed regions of interest (ROIs) to improve data quality for lesion identification.

- Designed a dual-stream framework integrating a pre-trained BERT and a 3D Vision Transformer (ViT); utilized contrastive learning and cross-attention mechanisms to align semantic text features with visual representations.
- Researched the role of federated learning and fairness metrics in protecting data privacy.

ACADEMIC SERVICES

- **Program Committee:** AAAI Conference on Artificial Intelligence (AAAI 2026)

INDUSTRY EXPERIENCES

- **Guangzhou Xiaopeng Motors Technology Co., Ltd.**Jun. 2024 - Aug. 2024
*Big Data Analytics Intern*Guangzhou, China
 - Fulfilled data collection and image screening duties for an autonomous parking project (AVM-BEV-FSD); established corner cases for parking scenarios to meet daily data quotas.
 - Executed view stitching of parking scenarios using the IPS model; utilized the Redash platform to monitor daily data collection, identify data anomalies, and analyze their root causes.
 - Identified and reported image quality issues such as scene repetition and IPS inconsistencies; monitored various data collection metrics to inform potential risks and adjust volumes for different parking scenarios.
- **Sun Yat-Sen Memorial Hospital, Sun Yat-Sen University**Jan. 2024 - Feb. 2024
*Data Analytics Intern*Guangzhou, China
 - Applied custom masks on 2D image slices to extract specific Regions of Interest (ROIs) for analysis.
 - Performed data compression and content extraction on medical imaging data (NIfTI format).

SKILLS

- **Programming Languages:** Python, JavaScript, Java, C++, Lua, Bash, HTML, CSS
- **Tools & Frameworks:** Pytorch, Matlab, Neovim, Linux, Latex, Git, Docker, CUDA